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SEMICONDUCTOR DEVICE AND METHOD FOR MANUFACTURING SEMICONDUCTOR DEVICE

Abstract

Provided is a semiconductor device including: a transistor portion; a current sensing portion which detects a current flowing through the transistor portion; a gate pad which is provided above a semiconductor substrate; and a resistance adjustment portion which is electrically connected to the gate pad and adjusts gate resistances of the transistor portion and the current sensing portion, in which the resistance adjustment portion includes a main adjustment portion which is electrically connected to a gate conductive portion of the transistor portion, and a sense adjustment portion which is electrically connected to a gate conductive portion of the current sensing portion, and each of the main adjustment portion and the sense adjustment portion includes a diode element portion including a plurality of diodes provided in anti-parallel and a resistance portion which is connected to the diode element portion.

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Background/Summary

[0001] The contents of the following patent application(s) are incorporated herein by reference:
[0002] NO. 2023-102280 filed in JP on Jun. 22, 2023 [0003] NO. PCT/JP2024/014745 filed in WO on Apr. 11, 2024.

BACKGROUND

1. Technical Field

[0004] The present invention relates to a semiconductor device and a method for manufacturing [0005] the semiconductor device.

2. Related Art

[0006] Patent Document 1 discloses a semiconductor device “including a main IGBT cell and a sense IGBT cell connected in parallel with each other”. [0007] Prior Art Documents [0008] Patent Documents [0009] Patent Document 1: Japanese Patent Application Publication No. 2019-149558 [0010] Patent Document 2: Japanese Patent No. 5591213 [0011] Patent Document 3: Japanese Patent No. 6102394

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] FIG. 1A illustrates an example of a top view of a semiconductor device **100**.
[0013] FIG. 1B illustrates an example of an enlarged view of an upper surface of the semiconductor device **100**.
[0014] FIG. 1C illustrates an example of a cross section a-a' in FIG. 1B.
[0015] FIG. 2A illustrates an example of a circuit configuration of the semiconductor device **100**. FIG. 2B illustrates a modification of the circuit configuration of the semiconductor device **100**.
[0016] FIG. 3A illustrates waveforms of a collector current Ic' and a sense current Ise' flowing through a semiconductor device of a comparative example.
[0017] FIG. 3B illustrates waveforms of a collector current Ic and a sense current Ise flowing through the semiconductor device **100**.
[0018] FIG. 4A illustrates an example of an enlarged view of the upper surface around a current sense pad **190** and a gate pad **150**.
[0019] FIG. 4B illustrates an example of a cross section b-b' in FIG. 4A.
[0020] FIG. 5A illustrates an example of a cross section c-c' in FIG. 4A.
[0021] FIG. 5B illustrates an example of a top view of a diode element portion **61**.
[0022] FIG. 5C illustrates an example of a top view of a resistance portion **62**.
[0023] FIG. 6A illustrates an example of a cross section d-d' in FIG. 4A.
[0024] FIG. 6B illustrates an example of a cross section e-e' in FIG. 4A.

[0025] FIG. 7A illustrates an example of a method for manufacturing the semiconductor device **100** including a resistance adjustment portion **60**.

[0026] FIG. 7B illustrates an example of a method for manufacturing the semiconductor device **100** including the resistance adjustment portion **60**.

[0027] FIG. **8** illustrates a modification of the cross section c-c' in FIG. **4A**.

[0028] FIG. **9A** illustrates a modification of the cross section c-c' in FIG. **4A**.

[0029] FIG. **9B** illustrates a modification of the cross section c-c' in FIG. **4A**.

[0030] FIG. **10** illustrates an example of a top view of the resistance adjustment portion **60**.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

[0031] Hereinafter, the present invention will be described through embodiments of the invention, but the following embodiments do not limit the invention according to claims. In addition, not all combinations of features described in the embodiments are essential to a solution of the invention.

[0032] Herein, one side of a semiconductor substrate in a direction parallel to a depth direction of the semiconductor substrate is referred to as “front” or “upper”, and an opposite side is referred to as “back” or “lower”. One surface of two principal surfaces of a substrate, a layer or another member is referred to as an upper surface, and another surface is referred to as a lower surface. A direction indicated by “front”, “upper”, “back”, or “lower” is not limited to a gravitational direction or a direction of implementing the semiconductor device.

[0033] In the present specification, technical matters may be described using orthogonal coordinate axes of an X axis, a Y axis, and a Z axis. The orthogonal coordinate axes merely specify relative positions of components, and do not limit a specific direction. For example, the Z axis is not limited to indicate a height direction with respect to the ground. Note that a +Z axis direction and a -Z axis direction are directions opposite to each other. When a Z axis direction is described without describing the signs, it means that the direction is parallel to a +Z axis and a -Z axis. As used herein, when seen from the +Z axis direction, it may also be referred to “top view”.

[0034] In the present specification, a case where a term such as “same” or “equal” is mentioned may include a case where an error due to a variation in manufacturing or the like is included. The error is, for example, within 10%.

[0035] In the present specification, a conductivity type of a doping region doped with impurities is described as a P type or an N type. Note that conductivity types of respective doping regions may be of opposite polarities, respectively. Further, in the present specification, a description of a P+ type or an N+ type means a higher doping concentration than that of the P type or the N type, and a description of a P- type or an N- type means a lower doping concentration than that of the P type or the N type.

[0036] In the present specification, the doping concentration indicates a concentration of an impurity activated as a donor or an acceptor. Herein, in some cases, a concentration difference between a donor and an acceptor may be a higher concentration of the two, the donor or the acceptor. The concentration difference can be measured by a capacitance-voltage profiling method (CV method). Also, a carrier concentration measured by Spread Resistance (SR) measurement method may be a concentration of a donor or an acceptor. In addition, in a case where a concentration distribution of a donor or an acceptor has a peak, the peak value may be a concentration of a donor or acceptor in the region. In a region where donors or acceptors are present, when concentration of the donors or the acceptors is substantially uniform or the like, an average value of donor concentration or acceptor concentration in this region may be set as donor concentration or acceptor concentration.

[0037] FIG. **1A** illustrates an example of a top view of a semiconductor device **100**. The semiconductor device **100** is a semiconductor chip including a transistor portion **70**. The semiconductor device **100** includes a temperature sensing portion **180**, and may be mounted on a module such as an intelligent power module (IPM).

[0038] The transistor portion **70** includes a transistor such as an insulated gate bipolar transistor

(IGBT). The transistor portion **70** may be another transistor such as a MOSFET. The semiconductor device **100** may be a reverse conducting IGBT (RC-IGBT) having a diode such as a free wheel diode (FWD) on a same chip.

[0039] A semiconductor substrate **10** may be a silicon substrate, may be a silicon carbide substrate, or may be a nitride semiconductor substrate such as gallium nitride, or the like. The semiconductor substrate **10** in the present example is the silicon substrate. The semiconductor substrate **10** includes an active portion **104** and an outer peripheral portion **106**. Note that when merely referred to as the top view in the present specification, it means that the semiconductor substrate **10** is viewed from an upper surface side. As will be described later, the semiconductor substrate **10** includes a front surface **21** and a back surface **23**.

[0040] The transistor portion **70** is a region obtained by projecting a collector region provided on a lower surface side of the semiconductor substrate **10** onto an upper surface of the semiconductor substrate **10**. The collector region is of a second conductivity type. The collector region is of P+type as an example.

[0041] Note that the transistor portion **70** in the present example has a trench portion extending in a Y axis direction. However, the transistor portion **70** may have a trench portion extending in an X axis direction.

[0042] The active portion **104** includes the transistor portion **70**. The active portion **104** is a region in which a main current flows between the upper and lower surfaces of the semiconductor substrate **10** when the semiconductor device **100** is controlled into an on state. In other words, the active portion is a region in which a current flows inside the semiconductor substrate **10** in a depth direction from the upper surface to the lower surface or from the lower surface to the upper surface of the semiconductor substrate **10**. In the present specification, the transistor portion **70** is referred to as an element portion or an element region.

[0043] Note that in top view, a region sandwiched between the two element portions is also the active portion **104**. In the present example, a region where a gate metal wiring **152** is provided while being sandwiched between the element portions is also included in the active portion **104**.

[0044] The gate metal wiring **152** is formed of a material containing metal. For example, the gate metal wiring **152** is formed of aluminum, an aluminum-silicon alloy, or an aluminum-silicon-copper alloy. The gate metal wiring **152** is electrically connected to a gate conductive portion of the transistor portion **70**, and supplies a gate voltage to the transistor portion **70**. The gate metal wiring **152** is provided so as to surround an outer periphery of the active portion **104** in top view. The gate metal wiring **152** is electrically connected to a gate pad **150** provided on the outer peripheral portion **106**.

[0045] A thickness and a width of the gate metal wiring **152** may not be uniform. The width of the gate metal wiring **152** may be formed to decrease as a distance from the gate pad **150** increases. The width of the gate metal wiring **152** may be formed such that, in a resistance adjustment portion to be described later, the width of the gate metal wiring **152** from the gate pad **150** to the resistance adjustment portion is larger than the width of the gate metal wiring **152** from the resistance adjustment portion to the transistor portion **70**. By changing the thickness and the width of the gate metal wiring **152**, a gate resistance in a current path from the gate pad **150** to the transistor portion **70** can be finely adjusted.

[0046] The outer peripheral portion **106** is a region between the active portion **104** and an end side **102** of the semiconductor substrate **10** in top view. In top view, the outer peripheral portion **106** is provided to enclose the active portion **104**. In the outer peripheral portion **106**, one or more metal pads for connecting the semiconductor device **100** and an external device by wires or the like may be arranged. Note that the outer peripheral portion **106** may include an edge termination structure portion. The edge termination structure portion reduces electric field strength on the upper surface side of the semiconductor substrate **10**. For example, the edge termination structure portion has a structure of a guard ring, a field plate, an RESURF, and a combination thereof.

[0047] A front surface side metal layer **110** is provided above the semiconductor substrate **10**. The front surface side metal layer **110** may include an emitter electrode **52**, an anode pad **130**, a cathode pad **140**, the gate pad **150**, and a current sense pad **190**. The front surface side metal layer **110** may be connected to an electrode outside semiconductor device **100** by wire bonding or the like. Note that, a number of at least one front surface side metal layer **110** and a position of the front surface side metal layer **110** are not limited to the present example.

[0048] The gate pad **150** is provided above the semiconductor substrate **10**. The gate pad **150** is electrically connected to the gate conductive portion of the transistor portion **70** via the gate metal wiring **152**. The gate pad **150** is set to a gate potential. The gate pad **150** in the present example has a rectangular shape in top view, but is not limited thereto.

[0049] The current sense pad **190** is electrically connected to a current sensing portion **90** to be described later. The current sense pad **190** detects a current flowing through the current sensing portion **90**. The current sense pad **190** in the present example has a rectangular shape in top view, but is not limited thereto.

[0050] The anode pad **130** is electrically connected to an anode of the temperature sensing portion **180**. The anode pad **130** is connected to the anode of the temperature sensing portion **180** by an anode wiring **132**. The anode pad **130** in the present example has a rectangular shape in top view, but is not limited thereto.

[0051] The cathode pad **140** is electrically connected to a cathode of the temperature sensing portion **180**. The cathode pad **140** is connected to the cathode of the temperature sensing portion **180** by a cathode wiring **142**. The cathode pad **140** in the present example has a rectangular shape in top view, but is not limited thereto.

[0052] The temperature sensing portion **180** is provided above the active portion **104**. The temperature sensing portion **180** senses a temperature of the active portion **104**. The temperature sensing portion **180** may include a diode formed of monocrystalline or polycrystalline silicon. The temperature sensing portion **180** is used for detecting a temperature of the semiconductor device **100** and protecting the semiconductor chip from overheating. The temperature sensing portion **180** is connected to a constant current source. When the temperature of the semiconductor device **100** changes, a forward voltage of a current flowing through the temperature sensing portion **180** changes. The semiconductor device **100** can detect the temperature based on the change in the forward voltage. The temperature sensing portion **180** has its long side along the Y axis direction and its short side along the X axis direction, but is not limited thereto.

[0053] The temperature sensing portion **180** in the present example is provided near a center of the active portion **104** in top view. The temperature sensing portion **180** is provided adjacent to the transistor portion **70**. The collector region of the second conductivity type may be provided on a back surface side of the semiconductor substrate **10** provided with the temperature sensing portion **180**.

[0054] The anode wiring **132** and the cathode wiring **142** are provided above the active portion **104** in top view. In addition, the anode wiring **132** and the cathode wiring **142** are provided to extend from the temperature sensing portion **180** to the outer peripheral portion **106**. The anode wiring **132** and the cathode wiring **142** in the present example are provided to extend from the temperature sensing portion **180** in the Y axis direction. The anode wiring **132** and the cathode wiring **142** may be made of a same material as that of the front surface side metal layer **110**.

[0055] FIG. **1B** illustrates an example of an enlarged view of an upper surface of the semiconductor device **100**. In the present example, an enlarged view of an end portion of the active portion **104** is illustrated.

[0056] The semiconductor device **100** in the present example includes, at the front surface **21** of the semiconductor substrate **10**, a gate trench portion **40**, a dummy trench portion **30**, an emitter region **12**, a base region **14**, a contact region **15**, and a well region **17**. In addition, the semiconductor device **100** in the present example includes the emitter electrode **52** and the gate metal wiring **152**

provided above the front surface **21** of the semiconductor substrate **10**. The gate trench portion **40** is an example of a MOS gate structure provided in the semiconductor device **100**.

[0057] The emitter electrode **52** is provided above the gate trench portion **40**, the dummy trench portion **30**, the emitter region **12**, the base region **14**, the contact region **15**, and the well region **17**. In addition, the gate metal wiring **152** is provided above a connection portion **25** and the well region **17**.

[0058] The emitter electrode **52** and the gate metal wiring **152** are formed of a material containing metal. At least a partial region of the emitter electrode **52** may be formed of metal such as aluminum (Al) or of a metal alloy such as an aluminum-silicon alloy (AlSi) or an aluminum-silicon-copper alloy (AlSiCu). At least a partial region of the gate metal wiring **152** may be formed of metal such as aluminum (Al) or a metal alloy such as an aluminum-silicon alloy (AlSi) or an aluminum-silicon-copper alloy (AlSiCu). The emitter electrode **52** and the gate metal wiring **152** may include a barrier metal layer formed of titanium, a titanium compound, or the like below a region formed of aluminum or the like. The barrier metal layer will be described later. The emitter electrode **52** and the gate metal wiring **152** are provided separately from each other.

[0059] The emitter electrode **52** and the gate metal wiring **152** are provided above the semiconductor substrate **10** with an interlayer dielectric film **38** interposed therebetween. The interlayer dielectric film **38** is omitted in FIG. **1B**. A contact hole **54**, a contact hole **55**, and a contact hole **56** are provided to penetrate the interlayer dielectric film **38**.

[0060] A contact hole **53** and the contact hole **55** electrically connect the gate metal wiring **152** and the gate conductive portion in the transistor portion **70** via the connection portion **25**. The contact hole **53** connects the gate conductive portion of the gate trench portion **40** and the connection portion **25**. An inside of the contact hole **53** may be filled with a same material as that of the connection portion **25**. The contact hole **55** connects the connection portion **25** and the gate metal wiring **152**. A contact portion formed of tungsten or the like may be formed inside the contact hole **55**. The contact portion will be described later.

[0061] The contact hole **56** connects the emitter electrode **52** with a dummy conductive portion inside the dummy trench portion **30**. A contact portion formed of tungsten or the like may be formed inside the contact hole **56**.

[0062] The connection portion **25** is connected to the front surface side metal layer **110** such as the emitter electrode **52** or the gate metal wiring **152**. In an example, the connection portion **25** is provided between the gate metal wiring **152** and the gate conductive portion. The connection portion **25** in the present example may be provided to extend in the X axis direction and electrically connected to the gate conductive portion. The connection portion **25** may also be provided between the emitter electrode **52** and the dummy conductive portion. In the present example, the connection portion **25** is not provided between the emitter electrode **52** and the dummy conductive portion. The connection portion **25** is formed of a conductive material such as polysilicon doped with impurities. The connection portion **25** in the present example is polysilicon (N+) doped with impurities of the N type. The connection portion **25** may be a polysilicon layer formed in a same step as that of a first polysilicon layer **91** or a second polysilicon layer **92** described later. The connection portion **25** may be provided above the front surface **21** of the semiconductor substrate **10** via a dielectric film such as an oxide film.

[0063] The gate trench portions **40** are examples of a plurality of trench portions extending in a predetermined extending direction on a front surface **21** side of the semiconductor substrate **10**.

[0064] The gate trench portions **40** are arrayed at a predetermined interval along a predetermined array direction (the X axis direction in the present example). The gate trench portion **40** in the present example may have two extending parts **41** which extend along an extending direction (the Y axis direction in the present example) parallel to the front surface **21** of the semiconductor substrate **10** and perpendicular to the array direction, and a connecting part **43** which connects the two extending parts **41**.

[0065] At least a part of the connecting part **43** is preferably formed in a curved shape. Connecting end portions of the two extending parts **41** of the gate trench portion **40** can reduce electric field strength at the end portions of the extending parts **41**. In the connecting part **43** of the gate trench portion **40**, the gate metal wiring **152** may be electrically connected to the gate conductive portion via the connection portion **25**.

[0066] The dummy trench portions **30** are examples of a plurality of trench portions extending in a predetermined extending direction on the front surface **21** side of the semiconductor substrate **10**. The dummy trench portion **30** is a trench portion which is electrically connected to the emitter electrode **52**. Similarly to the gate trench portions **40**, the dummy trench portions **30** are arrayed at a predetermined interval along a predetermined array direction (the X axis direction in the present example). Although the dummy trench portion **30** in the present example has an I shape on the front surface **21** of the semiconductor substrate **10**, the dummy trench portion **30** may have a U shape on the front surface **21** of the semiconductor substrate **10**, similarly to the gate trench portion **40**. That is, the dummy trench portion **30** may include two extending parts extending along the extending direction and a connecting part connecting the two extending parts.

[0067] The transistor portion **70** in the present example has a structure in which two gate trench portions **40** and two dummy trench portions **30** are repetitively arrayed. That is, the transistor portion **70** in the present example has the gate trench portions **40** and the dummy trench portions **30** at a ratio of 1:1. For example, the transistor portion **70** has one dummy trench portion **30** between two extending parts **41**.

[0068] Note that the ratio between the gate trench portions **40** and the dummy trench portions **30** is not limited to that in the present example. The ratio of the gate trench portions **40** may be larger than a ratio of the dummy trench portions **30**, or the ratio of the dummy trench portions **30** may be larger than the ratio of the gate trench portions **40**. A ratio between the gate trench portions **40** and the dummy trench portions **30** may be 2:3, or may be 2:4. In addition, the transistor portion **70** may not include the dummy trench portions **30** with all trench portions being the gate trench portions **40**.

[0069] The well region **17** is a region of the second conductivity type which is provided on the front surface **21** side of the semiconductor substrate **10** relative to a drift region **18** to be described later. The well region **17** is an example of the well region provided in a peripheral side of the active portion **104**. The well region **17** is of the P+ type as an example. The well region **17** is formed in a predetermined range from the end portion of the active region on a side where the gate metal wiring **152** is provided. A diffusion depth of the well region **17** may be deeper than depths of the gate trench portion **40** and the dummy trench portion **30**. Partial regions of the gate trench portions **40** and the dummy trench portions **30** at the gate metal wiring **152** side are formed in the well region **17**. Bottoms of ends in the extending direction of the gate trench portion **40** and the dummy trench portion **30** may be covered with the well region **17**.

[0070] The contact hole **54** is formed above each region of the emitter region **12** and the contact region **15** in the transistor portion **70**. The contact hole **54** is not provided above the well regions **17** provided at both ends in the Y axis direction. In this manner, one or more contact holes **54** are formed in the interlayer dielectric film. The one or more contact holes **54** may be provided to extend in the extending direction.

[0071] A mesa portion **71** is a mesa portion provided adjacent to the trench portion in a plane parallel to the front surface **21** of the semiconductor substrate **10**. The mesa portion may be a part of the semiconductor substrate **10** sandwiched between two adjacent trench portions, and may be a part from the front surface **21** of the semiconductor substrate **10** to a depth of a lowermost bottom portion of each trench portion. An extending part of each trench portion may be defined as one trench portion. That is, a region sandwiched between two extending parts may be defined as a mesa portion.

[0072] The mesa portion **71** is provided adjacent to at least one of the dummy trench portion **30** or

the gate trench portion **40** in the transistor portion **70**. The mesa portion **71** has the well region **17**, the emitter region **12**, the base region **14**, and the contact region **15** at the front surface **21** of the semiconductor substrate **10**. In the mesa portion **71**, emitter regions **12** and contact regions **15** are alternately provided in the extending direction.

[0073] The base region **14** is a region of the second conductivity type which is provided on the front surface **21** side of the semiconductor substrate **10**. The base region **14** is of the P- type as an example. The base regions **14** may be provided at both end portions of the mesa portion **71** in the Y axis direction at the front surface **21** of the semiconductor substrate **10**. Note that FIG. **1B** illustrates only one end portion of the base region **14** in the Y axis direction.

[0074] The emitter region **12** is a region of a first conductivity type which has a doping concentration higher than that of the drift region **18**. The emitter region **12** in the present example is of the N+ type as an example. Examples of a dopant of the emitter region **12** include arsenic (As). The emitter region **12** is provided in contact with the gate trench portion **40** at the front surface **21** in the mesa portion **71**. The emitter region **12** may be provided to extend in the X axis direction from one to another of two trench portions sandwiching the mesa portion **71**. The emitter region **12** is also provided below the contact hole **54**.

[0075] In addition, the emitter region **12** may be or may not be in contact with the dummy trench portion **30**. The emitter region **12** in the present example is in contact with the dummy trench portion **30**.

[0076] The contact region **15** is a region of the second conductivity type which is provided above the base region **14** and has a doping concentration higher than that of the base region **14**. The contact region **15** in the present example is of the P+ type as an example. The contact region **15** in the present example is provided at the front surface **21** in the mesa portion **71**. The contact region **15** may be provided in the X axis direction from one to another of the two trench portions sandwiching the mesa portion **71**. The contact region **15** may be or may not be in contact with the gate trench portion **40** or the dummy trench portion **30**. The contact region **15** in the present example is in contact with the dummy trench portion **30** and the gate trench portion **40**. The contact region **15** is also provided below the contact hole **54**.

[0077] FIG. **1C** illustrates an example of a cross section a-a' in FIG. **1B**. The cross section a-a' is an XZ plane passing through the emitter region **12** in the transistor portion **70**. In the cross section a-a', the semiconductor device **100** in the present example includes the semiconductor substrate **10**, the interlayer dielectric film **38**, the emitter electrode **52**, and a collector electrode **24**. The collector electrode **24** is an example of a back surface side metal layer provided in contact with the back surface **23** of the semiconductor substrate **10**. The emitter electrode **52** is formed above the semiconductor substrate **10** and the interlayer dielectric film **38**.

[0078] The drift region **18** is a region of the first conductivity type which is provided in the semiconductor substrate **10**. The drift region **18** in the present example is of the N- type as an example. The drift region **18** may be a region which has remained without other doping regions formed in the semiconductor substrate **10**. That is, a doping concentration of the drift region **18** may be a doping concentration of the semiconductor substrate **10**.

[0079] A buffer region **20** is a region of the first conductivity type which is provided on a back surface **23** side of the semiconductor substrate **10** relative to the drift region **18**. The buffer region **20** in the present example is of the N type as an example. A doping concentration of the buffer region **20** is higher than the doping concentration of the drift region **18**. The buffer region **20** may function as a field stop layer which prevents a depletion layer extending from a lower surface side of the base region **14** from reaching the collector region **22** of the second conductivity type. Note that the buffer region **20** may be omitted.

[0080] The collector region **22** is provided below the buffer region **20** in the transistor portion **70**. The collector region **22** is of the second conductivity type. The collector region **22** in the present example is of the P+ type as an example.

[0081] The collector electrode **24** is formed at the back surface **23** of the semiconductor substrate **10**. The collector electrode **24** is formed of a conductive material such as metal. The material of the collector electrode **24** may be a same as or different from the material of the emitter electrode **52**.

[0082] The base region **14** is a region of the second conductivity type which is provided above the drift region **18**. The base region **14** is provided in contact with the gate trench portion **40**. The base region **14** may be provided in contact with the dummy trench portion **30**.

[0083] The emitter region **12** is provided above the base region **14**. The emitter region **12** is provided between the base region **14** and the front surface **21**. The emitter region **12** is provided in contact with the gate trench portion **40**. The emitter region **12** may be or may not be in contact with the dummy trench portion **30**.

[0084] An accumulation region **16** is a region of the first conductivity type which is provided on the front surface **21** side of the semiconductor substrate **10** relative to the drift region **18**. The accumulation region **16** in the present example is of the N⁺ type as an example. Noted that the accumulation region **16** may not be provided.

[0085] The accumulation region **16** is provided in contact with the gate trench portion **40**. The accumulation region **16** may or may not be in contact with the dummy trench portion **30**. A doping concentration of the accumulation region **16** is higher than the doping concentration of the drift region **18**. An ion implantation dose amount of the accumulation region **16** may be 1.0E+12 cm.sup.-2 or more and 1.0E+13 cm.sup.-2 or less. Alternatively, the ion implantation dose amount of the accumulation region **16** may be 3.0E+12 cm.sup.-2 or more and 6.0E+12 cm.sup.-2 or less. Providing the accumulation region **16** can increase a carrier injection enhancement effect (IE effect) to reduce an on-voltage of the transistor portion **70**.

[0086] One or more gate trench portions **40** and one or more dummy trench portions **30** are provided at the front surface **21**. Each trench portion is provided from the front surface **21** to the drift region **18**. In a region provided with at least any of the emitter region **12**, the base region **14**, the contact region **15**, or the accumulation region **16**, each trench portion also penetrates these regions to reach the drift region **18**. A configuration in which the trench portion penetrates the doping region is not limited to a configuration which is manufactured in an order of forming the doping region and then forming the trench portion. The configuration in which the trench portion penetrates the doping region includes a configuration in which the trench portions are formed and then the doping region is formed between the trench portions.

[0087] The gate trench portion **40** has a gate trench, a gate dielectric film **42**, and a gate conductive portion **44** which are formed at the front surface **21**. The gate dielectric film **42** is formed to cover an inner wall of the gate trench. The gate dielectric film **42** may be formed by oxidizing or nitriding a semiconductor at the inner wall of the gate trench. The gate conductive portion **44** is formed farther inward than the gate dielectric film **42** inside the gate trench. The gate dielectric film **42** insulates the gate conductive portion **44** from the semiconductor substrate **10**. The gate conductive portion **44** is formed of a conductive material such as polysilicon. The gate trench portion **40** is covered with the interlayer dielectric film **38** on the front surface **21**.

[0088] The gate conductive portion **44** includes a region opposing the adjacent base region **14** on a mesa portion **71** side with the gate dielectric film **42** interposed therebetween, in the depth direction of the semiconductor substrate **10**. When a predetermined voltage is applied to the gate conductive portion **44**, a channel is formed by an electron inversion layer in a surface layer of the base region **14** at an interface in contact with the gate trench.

[0089] The dummy trench portion **30** may have a same structure as that of the gate trench portion **40**. The dummy trench portion **30** has a dummy trench, a dummy dielectric film **32**, and a dummy conductive portion **34** which are formed on the front surface **21** side. The dummy dielectric film **32** is formed to cover an inner wall of the dummy trench. The dummy conductive portion **34** is formed inside the dummy trench, and is formed farther inward than the dummy dielectric film **32**. The dummy dielectric film **32** insulates the dummy conductive portion **34** from the semiconductor

substrate **10**. The dummy trench portion **30** may be covered with the interlayer dielectric film **38** on the front surface **21**.

[0090] The interlayer dielectric film **38** is provided above the semiconductor substrate **10**. The interlayer dielectric film **38** in the present example is provided in contact with the front surface **21**.

[0091] The emitter electrode **52** is provided above the interlayer dielectric film **38**. The interlayer dielectric film **38** is provided with one or more contact holes **54** for electrically connecting the emitter electrode **52** to the semiconductor substrate **10**. Similarly, the contact hole **53**, the contact hole **55**, and the contact hole **56** may be provided to penetrate the interlayer dielectric film **38**. A film thickness of the interlayer dielectric film **38** may be 0.8 μm or more and 1.2 μm or less.

[0092] The interlayer dielectric film **38** may be a silicon oxide film. The interlayer dielectric film **38** may be a boro-phospho silicate glass (BPSG) film, may be a borosilicate glass (BSG) film, or may be a phosphosilicate glass (PSG) film. The interlayer dielectric film **38** may also include a high temperature silicon oxide (HTO: High Temperature Oxide) film.

[0093] A back surface side lifetime control region **120** may be provided in the transistor portion **70**. However, the back surface side lifetime control region **120** may be omitted. The back surface side lifetime control region **120** is a region where a lifetime killer has intentionally been formed by implanting impurities inside the semiconductor substrate **10**, or the like. As an example, the back surface side lifetime control region **120** is formed by implanting helium into the semiconductor substrate **10**. The back surface side lifetime control region **120** may also be formed by implanting protons. By providing the back surface side lifetime control region **120**, a turn-off time can be reduced, and by suppressing a tail current, losses during switching can be reduced.

[0094] The lifetime killer is a recombination center of carriers. The lifetime killer may be a lattice defect. For example, the lifetime killer may be a vacancy, a divacancy, a defect complex of these with elements constituting the semiconductor substrate **10**, or dislocation. Furthermore, the lifetime killer may be a noble gas element such as helium and neon, a metal element such as platinum, or the like. An electron beam or a proton may be used for forming the lattice defect.

[0095] A lifetime killer concentration is a concentration at the recombination center of carriers. The lifetime killer concentration may be a concentration of the lattice defect. For example, the lifetime killer concentration may be a vacancy concentration of a vacancy, a divacancy, or the like, may be a defect complex concentration of these vacancies with elements constituting the semiconductor substrate **10**, or may be a dislocation concentration. Alternatively, the lifetime killer concentration may be a chemical concentration of the noble gas element such as helium and neon, or may be a chemical concentration of the metal element such as platinum.

[0096] The back surface side lifetime control region **120** is provided on the back surface **23** side relative to the center of the semiconductor substrate **10** in the depth direction of the semiconductor substrate **10**. The back surface side lifetime control region **120** in the present example is provided in the buffer region **20**. The back surface side lifetime control region **120** may be provided on an entire surface of the semiconductor substrate **10** in an XY plane by being formed without using a mask. The back surface side lifetime control region **120** may be provided on a part of the semiconductor substrate **10** in the XY plane by being formed using a mask. An impurity dose amount for forming the back surface side lifetime control region **120** may be $0.5\text{E}+10\text{ cm.sup.-2}$ or more and $1.0\text{E}+14\text{ cm.sup.-2}$ or less, or may be $5.0\text{E}+10\text{ cm.sup.-2}$ or more and $1.0\text{E}+13\text{ cm.sup.-2}$ or less.

[0097] The back surface side lifetime control region **120** may be formed by an implantation from the back surface **23** side. Accordingly, it becomes easy to avoid an effect on the front surface **21** side of the semiconductor device **100**. For example, the back surface side lifetime control region **120** is formed by radiating helium or protons from the back surface **23** side. Herein, which of the front surface **21** side and the back surface **23** side the implantation is performed from for forming the back surface side lifetime control region **120** can be determined by acquiring a state of the front surface **21** side by an SR method or a measurement of a leakage current.

[0098] FIG. 2A illustrates an example of a circuit configuration of the semiconductor device **100**. The semiconductor device **100** in the present example includes a resistance adjustment portion **60**. In the semiconductor device **100**, the transistor portion **70** and the current sensing portion **90** are connected in parallel to each other.

[0099] The resistance adjustment portion **60** is electrically connected to the gate pad **150**, and adjusts gate resistances of the transistor portion **70** and the current sensing portion **90**. The resistance adjustment portion **60** includes a main adjustment portion **160** and a sense adjustment portion **260**.

[0100] The main adjustment portion **160** is electrically connected to the gate conductive portion **44** of the transistor portion **70**. The sense adjustment portion **260** is electrically connected to a gate conductive portion **194** of the current sensing portion **90**. The gate conductive portion **194** will be described later. Each of the main adjustment portion **160** and the sense adjustment portion **260** includes a diode element portion **61** and a resistance portion **62**.

[0101] The diode element portion **61** includes a plurality of diodes provided in anti-parallel. Since the diode element portion **61** includes the plurality of diodes provided in anti-parallel, a gate wiring for turning on gates of the transistor portion **70** and the current sensing portion **90** and a gate wiring for turning off the gates can be set to different paths. Accordingly, different resistance values can be set between on and off of the gates.

[0102] The resistance portion **62** is connected to the diode element portion **61**. The resistance portion **62** has a resistance value such that the transistor portion **70** is turned on before the current sensing portion **90**, and the transistor portion **70** is turned off later than the current sensing portion **90**. The resistance value of the resistance portion **62** may be set in consideration of a built-in gate resistance of the transistor portion **70** and a built-in gate resistance of the current sensing portion **90**. The built-in gate resistance may be a parasitic gate resistance generated by a gate wiring resistor or the like.

[0103] The main adjustment portion **160** includes a main on-side diode **161** and a main off-side diode **162** as the diode element portion **61**. The main on-side diode **161** and the main off-side diode **162** are connected in anti-parallel between the gate pad **150** and the gate conductive portion **44**. The main adjustment portion **160** includes a main off-side resistor **167** as the resistance portion **62**.

[0104] The main on-side diode **161** has an anode electrically connected to the gate pad **150** and a cathode electrically connected to the gate conductive portion **44** of the transistor portion **70**. The main off-side diode **162** has an anode electrically connected to the gate conductive portion **44** of the transistor portion **70** and a cathode electrically connected to the gate pad **150**.

[0105] The main off-side resistor **167** is connected in series with the main off-side diode **162** between the gate pad **150** and the gate conductive portion **44**. The main off-side resistor **167** in the present example is electrically connected between the anode of the main off-side diode **162** and the gate conductive portion **44** of the transistor portion **70**. The main off-side resistor **167** may be connected between the cathode of the main off-side diode **162** and the gate pad **150**. The main off-side resistor **167** may be set such that an off operation of the transistor portion **70** is later than an off operation of the current sensing portion **90**.

[0106] The main adjustment portion **160** in the present example adjusts only the gate resistance on the off side and does not adjust the gate resistance on the on side. However, the main adjustment portion **160** may adjust the gate resistances on both the on side and the off side, or may adjust only the gate resistance on the on side.

[0107] The sense adjustment portion **260** includes a sense on-side diode **261** and a sense off-side diode **262** as the diode element portion **61**. The sense on-side diode **261** and the sense off-side diode **262** are connected in anti-parallel between the gate pad **150** and the gate conductive portion **194**. The sense adjustment portion **260** includes a sense on-side resistor **266** as the resistance portion **62**.

[0108] The sense on-side diode **261** has an anode electrically connected to the gate pad **150** and a

cathode electrically connected to the gate conductive portion **194** of the current sensing portion **90**. The sense off-side diode **262** has an anode electrically connected to the gate conductive portion **194** of the current sensing portion **90** and a cathode electrically connected to the gate pad **150**.

[0109] The sense on-side resistor **266** is connected in series with the sense on-side diode **261** between the gate pad **150** and the gate conductive portion **194**. The sense on-side resistor **266** in the present example is electrically connected between the cathode of the sense on-side diode **261** and the gate conductive portion **194** of the current sensing portion **90**. The sense on-side resistor **266** may be connected between the anode of the sense on-side diode **261** and the gate pad **150**. The sense on-side resistor **266** may be set such that an on operation of the current sensing portion **90** is later than an on operation of the transistor portion **70**.

[0110] The sense adjustment portion **260** in the present example adjusts only the gate resistance on the on side and does not adjust the gate resistance on the off side. However, the sense adjustment portion **260** may adjust the gate resistance on both the on side and the off side, or may adjust only the gate resistance on the off side.

[0111] Since the semiconductor device **100** in the present example includes the resistance adjustment portion **60**, switching speeds of the transistor portion **70** and the current sensing portion **90** can be adjusted. Accordingly, the semiconductor device **100** can suppress gate oscillation in the chip.

[0112] FIG. 2B illustrates a modification of the circuit configuration of the semiconductor device **100**. The semiconductor device **100** in the present example is different from the example of FIG. 2A in including a main on-side resistor **166** and a sense off-side resistor **267**. In the present example, differences from the example of FIG. 2A will be particularly described.

[0113] The main on-side resistor **166** is electrically connected between the cathode of the main on-side diode **161** and the gate conductive portion **44** of the transistor portion **70**. That is, the main on-side resistor **166** may be connected in series with the main on-side diode **161** and connected in parallel with the main off-side resistor **167** between the gate pad **150** and the gate conductive portion **44**. The main on-side resistor **166** may be electrically connected between the anode of the main on-side diode **161** and the gate pad **150**.

[0114] The sense off-side resistor **267** is electrically connected between the anode of the sense off-side diode **262** and the gate conductive portion **194** of the current sensing portion **90**. That is, the sense off-side resistor **267** may be connected in series with the sense off-side diode **262** and connected in parallel with the sense on-side resistor **266** between the gate pad **150** and the gate conductive portion **194**. The sense off-side resistor **267** may be connected between the cathode of the sense off-side diode **262** and the gate pad **150**.

[0115] In the semiconductor device **100** in the present example, the switching speed during the on operation of the transistor portion **70** and the switching speed during the off operation of the current sensing portion **90** can be further adjusted. That is, the semiconductor device **100** in the present example can adjust the switching speed during the on/off operation of the transistor portion **70** and adjust the switching speed during the on/off operation of the current sensing portion **90**.

[0116] The resistance adjustment portion **60** has a resistance value such that the transistor portion **70** is turned on before the current sensing portion **90**, and the transistor portion **70** is turned off later than the current sensing portion **90**. A resistance value of the main on-side resistor **166** may be smaller than a resistance value of the sense on-side resistor **266**. The resistance value of the main on-side resistor **166** may be 5 times or more and 100 times or less the resistance value of the sense on-side resistor **266**. A resistance value of the main off-side resistor **167** may be larger than a resistance value of the sense off-side resistor **267**. The resistance value of the main off-side resistor **167** may be 1 time or more and 10 times or less the resistance value of the sense off-side resistor **267**.

[0117] FIG. 3A illustrates waveforms of a collector current $I_{c'}$ and a sense current $I_{se'}$ flowing through a semiconductor device of a comparative example. In the comparative example, when a

gate voltage V_{ge} is turned off, the sense current I_{se} is turned off later than the collector current I_c . In this manner, when turn-off of a sensing portion is later than turn-off of a main transistor portion, a current flows into the sensing portion, and the current sensing portion may be broken or an oscillation phenomenon may occur. Similarly, when turn-on of the sensing portion is earlier than turn-on of the main transistor portion, a current flows into the sensing portion, and the current sensing portion may be broken or an oscillation phenomenon may occur.

[0118] FIG. 3B illustrates waveforms of a collector current I_c and a sense current I_{se} flowing through the semiconductor device **100**. In the present example, when the gate voltage V_{ge} is turned off, the sense current I_{se} of the current sensing portion **90** is turned off before the collector current I_c of the transistor portion **70**. In this manner, the semiconductor device **100** can avoid breakdown of the current sensing portion **90** by adjusting the switching speed such that turn-off of the current sensing portion **90** is earlier than turn-off of the transistor portion **70**. Similarly, the semiconductor device **100** may avoid breakdown of the current sensing portion **90** by adjusting the switching speed such that turn-on of the current sensing portion **90** occurs at a same time as or later than turn-on of the transistor portion **70**.

[0119] FIG. 4A illustrates an example of an enlarged view of the upper surface around the current sense pad **190** and the gate pad **150**. In the transistor portion **70** in the present example, all trench portions are the gate trench portions **40**, but the transistor portion **70** may include the dummy trench portion **30**. The structure of the transistor portion **70** may be the same as that of the semiconductor device **100** in FIG. 1B. The semiconductor device **100** includes the current sensing portion **90** and the resistance adjustment portion **60**.

[0120] The current sensing portion **90** detects a current flowing through the transistor portion **70**. A trench structure of the current sensing portion **90** is provided below the current sense pad **190**. The current sensing portion **90** has a structure corresponding to the transistor portion **70**, and simulates the operation of the transistor portion **70**. Since a current proportional to the current flowing through the transistor portion **70** flows through the current sensing portion **90**, the current flowing through the transistor portion **70** can be monitored by using the current sensing portion **90**.

Accordingly, it is possible to protect the transistor portion **70** from overcurrent.

[0121] The gate pad **150** may be provided in common between the transistor portion **70** and the current sensing portion **90**. That is, a common gate control signal may be input to the transistor portion **70** and the current sensing portion **90**. The gate pad **150** may be provided in common between the main adjustment portion **160** and the sense adjustment portion **260**. The gate pad **150** is connected to the main adjustment portion **160** and the sense adjustment portion **260** by the gate metal wiring **152**.

[0122] The main adjustment portion **160** may be provided adjacent to the transistor portion **70**. The main adjustment portion **160** in the present example is provided further away from the gate pad **150** than the current sense pad **190**, but may be provided closer to the gate pad **150** than the current sense pad **190**. The main adjustment portion **160** may be connected to the transistor portion **70** by using at least one of the connection portion **25** or the gate metal wiring **152**. The main adjustment portion **160** in the present example is connected to the transistor portion **70** by the connection portion **25**.

[0123] The main on-side diode **161** and the main off-side diode **162** may be arranged adjacent to each other. Being arranged adjacent to each other may mean that the gate metal wiring **152** is not present between the main on-side diode **161** and the main off-side diode **162**. The main on-side diode **161** and the main off-side diode **162** in the present example are arranged side by side in the extending direction of the trench, but are not limited thereto.

[0124] The sense adjustment portion **260** may be provided adjacent to the current sense pad **190**. That is, the gate metal wiring **152** may not be provided between the sense adjustment portion **260** and the current sense pad **190**. The sense adjustment portion **260** in the present example is provided between the current sense pad **190** and the gate pad **150**, but is not limited thereto. The sense

adjustment portion **260** may be connected to the current sensing portion **90** by using at least one of the connection portion **25** or the gate metal wiring **152**. The sense adjustment portion **260** in the present example is connected to the current sensing portion **90** by the connection portion **25**.

[0125] The sense on-side diode **261** and the sense off-side diode **262** may be arranged adjacent to each other. That is, the gate metal wiring **152** may not be provided between the sense on-side diode **261** and the sense off-side diode **262**. The sense on-side diode **261** and the sense off-side diode **262** in the present example are arranged side by side in the extending direction of the trench, but are not limited thereto.

[0126] The gate metal wiring **152** may connect the gate pad **150** and the resistance adjustment portion **60**. The gate metal wiring **152** in the present example connects each of the main on-side diode **161**, the main off-side diode **162**, the sense on-side diode **261**, and the sense off-side diode **262** to the gate pad **150**. The gate metal wiring **152** may be connected to the main adjustment portion **160** by using a third contact portion **173** provided in the contact hole **55**.

[0127] The resistance adjustment portion **60** may have a stacked structure in which the diode element portion **61** and the resistance portion **62** are stacked. By stacking the diode element portion **61** and the resistance portion **62** to increase an area of the resistance adjustment portion **60**, an ESD breakdown tolerance can be improved. The main on-side diode **161** may be provided to be stacked with the main on-side resistor **166**. The main off-side diode **162** may be provided to be stacked with the main off-side resistor **167**. The sense on-side diode **261** may be provided to be stacked with the sense on-side resistor **266**. The sense off-side diode **262** may be provided to be stacked with the sense off-side resistor **267**.

[0128] Here, the switching speeds of the transistor portion **70** and the current sensing portion **90** may change according to areas and the like thereof. The area of the current sensing portion **90** may be restricted depending on a chip size, an arrangement position of the pad, and the like. In the semiconductor device **100** in the present example, the switching speed can be adjusted according to a capacitance ratio based on the areas and the like of the transistor portion **70** and the current sensing portion **90**, and thus it is possible to suppress the gate oscillation in the chip regardless of a chip layout.

[0129] FIG. **4B** illustrates an example of a cross section b-b' in FIG. **4A**. The cross section b-b' is an XZ plane passing through the emitter region **12** in the current sensing portion **90**. The semiconductor device **100** in the present example includes the semiconductor substrate **10**, the interlayer dielectric film **38**, the current sense pad **190**, and the collector electrode **24** in the cross section b-b'. The current sense pad **190** is formed above the semiconductor substrate **10** and the interlayer dielectric film **38**. The current sensing portion **90** may include the contact region **15** similarly to the transistor portion **70**.

[0130] FIG. **5A** illustrates an example of a cross section c-c' in FIG. **4A**. The resistance adjustment portion **60** in the present example has a stacked structure in which the diode element portion **61** and the resistance portion **62** are stacked. The cross section c-c' is a YZ plane passing through a first contact portion **171** and a second contact portion **172** in the resistance adjustment portion **60**. The cross section c-c' shows a region of the sense on-side diode **261** in the resistance adjustment portion **60** in FIG. **4A**, but the same may apply to cross sections of the main on-side diode **161**, the main off-side diode **162**, and the sense off-side diode **262**.

[0131] The diode element portion **61** is provided above the resistance portion **62**. The diode element portion **61** in the present example is provided above the resistance portion **62** with a second insulating portion **82** interposed therebetween. The diode element portion **61** may be provided below the resistance portion **62**. The diode element portion **61** may be any one of the main on-side diode **161**, the main off-side diode **162**, the sense on-side diode **261**, or the sense off-side diode **262**. The resistance portion **62** may be any of the main on-side resistor **166**, the main off-side resistor **167**, the sense on-side resistor **266**, or the sense off-side resistor **267**.

[0132] The first contact portion **171** is provided to connect the front surface side metal layer **110**

and the diode element portion **61**. The first contact portion **171** in the present example connects the gate metal wiring **152** and the diode element portion **61**. The first contact portion **171** is provided to extend from an upper end of the interlayer dielectric film **38** to an upper end of a first insulating portion **81** through the diode element portion **61**, the second insulating portion **82**, and the resistance portion **62**. A material of the first contact portion **171** may be the same as that of the front surface side metal layer **110**. The first contact portion **171** may include a plug member such as tungsten and may include a barrier metal. The first contact portion **171** is provided inside a first opening **271**.

[0133] The second contact portion **172** is provided to connect the front surface side metal layer **110** and the resistance portion **62**. The second contact portion **172** in the present example connects the diode element portion **61** and the resistance portion **62**. The second contact portion **172** is provided to extend from the upper end of the interlayer dielectric film **38** to the upper end of the first insulating portion **81** through the diode element portion **61**, the second insulating portion **82**, and the resistance portion **62**. A material of the second contact portion **172** may be the same as that of the front surface side metal layer **110**. The second contact portion **172** may include a plug member such as tungsten and may include a barrier metal. The second contact portion **172** is provided inside a second opening **272**.

[0134] The first opening **271** and the second opening **272** may be formed by a common etching process. That is, depths of the first opening **271** and the second opening **272** may be the same. The first opening **271** and the second opening **272** may be formed by a same etching process as that of the contact hole of the transistor portion **70**. For example, the first opening **271** and the second opening **272** are formed by a same etching process as the contact hole **54**, the contact hole **55**, and the contact hole **56**.

[0135] The first insulating portion **81** is provided above the semiconductor substrate **10**. The first insulating portion **81** may be an oxide film such as an HTO film. The resistance portion **62** in the present example is provided on the first insulating portion **81**.

[0136] The second insulating portion **82** is provided above the first insulating portion **81**. The second insulating portion **82** may be an oxide film such as an HTO film. The second insulating portion **82** may be used to separate the diode element portion **61** and the resistance portion **62** which are stacked.

[0137] The interlayer dielectric film **38** may be provided above the first insulating portion **81** and the second insulating portion **82**. The interlayer dielectric film **38** may be formed by a process common with that of the interlayer dielectric film **38** of the transistor portion **70** and the current sensing portion **90**, or may be formed by a process different therefrom. The thickness of the interlayer dielectric film **38** may be larger than a thickness of the first insulating portion **81** and may be larger than a thickness of the second insulating portion **82**. The gate metal wiring **152** may be provided on the interlayer dielectric film **38**.

[0138] A third insulating portion **83** is provided around the first contact portion **171** and electrically separates the first contact portion **171** from the resistance portion **62**. The third insulating portion **83** may be formed by a same process as that of the second insulating portion **82**. A material of the third insulating portion **83** may be the same as a material of the second insulating portion **82**.

[0139] The diode element portion **61** and the resistance portion **62** may be formed of polysilicon. The diode element portion **61** may be formed in the second polysilicon layer **92**, and the resistance portion **62** may be formed in the first polysilicon layer **91**.

[0140] The first polysilicon layer **91** is provided on the first insulating portion **81**. The first polysilicon layer **91** may be a polysilicon layer formed by a process common with that of at least one of the gate conductive portion **44** or the gate conductive portion **194**. The first polysilicon layer **91** may be polysilicon formed by a process common with that of the connection portion **25**. The polysilicon formed by the common process may be polysilicon formed simultaneously by using a same mask. A thickness of the first polysilicon layer **91** may be 0.5 μm or more and 1 μm or less.

[0141] The second polysilicon layer **92** is provided on the second insulating portion **82**. The second polysilicon layer **92** may be polysilicon formed by a process common with that of a temperature sense diode of the temperature sensing portion **180**. A thickness of the second polysilicon layer **92** may be smaller than the thickness of the first polysilicon layer **91**. That is, a thickness of the resistance portion **62** may be smaller than a thickness of the diode element portion **61**. The thickness of the second polysilicon layer **92** may be 0.3 μm or more and 0.7 μm or less.

[0142] FIG. 5B illustrates an example of a top view of the diode element portion **61**. The diode element portion **61** includes a P type region and an N type region provided in the second polysilicon layer **92**. The diode element portion **61** in the present example has one PN junction, but may have a plurality of PN junctions connected in series. The PN junction of the diode element portion **61** may be formed by ion implantation. Each of the P type region and the N type region may be formed by ion implantation. The P type region and the N type region may be formed by depositing a polysilicon layer of one conductivity type of the P type or the N type and then performing ion implantation of another conductivity type.

[0143] A structure of the diode element portion **61** in the present example may be a structure of at least one of the main on-side diode **161**, the main off-side diode **162**, the sense on-side diode **261**, or the sense off-side diode **262**. Structures of the main on-side diode **161**, the main off-side diode **162**, the sense on-side diode **261**, and the sense off-side diode **262** may be common or may be different from each other.

[0144] FIG. 5C illustrates an example of a top view of the resistance portion **62**. The resistance portion **62** includes the first polysilicon layer **91** connected to the second contact portion **172**. The resistance portion **62** in the present example has a resistance region **63** for adjusting the gate resistance. In the resistance portion **62**, the resistance may be adjusted by a shape of the first polysilicon layer **91**, or the resistance may be adjusted by a doping concentration of the first polysilicon layer **91**. The resistance region **63** in the present example includes a region in which the first polysilicon layer **91** meanders on the XY plane. The first polysilicon layer **91** may be the P type or the N type.

[0145] FIG. 6A illustrates an example of a cross section d-d' in FIG. 4A. The cross section d-d' is an XZ plane passing through the connection portion **25** for connecting the resistance adjustment portion **60** and the transistor portion **70**. The cross section d-d' passes through the first contact portion **171** and the third contact portion **173**.

[0146] The diode element portion **61** may or may not overlap the connection portion **25** in top view. The diode element portion **61** in the present example overlaps the connection portion **25** and is also provided above the connection portion **25**. The diode element portion **61** and the connection portion **25** may be separated by the second insulating portion **82**. The connection portion **25** may be connected to the gate metal wiring **152** via the third contact portion **173**.

[0147] The third contact portion **173** is used to connect the gate metal wiring **152** and the connection portion **25**. The third contact portion **173** is provided to extend from the upper end of the interlayer dielectric film **38** to the upper end of the first insulating portion **81** through the connection portion **25**. The third contact portion **173** may terminate at an upper surface of the connection portion **25** without penetrating the connection portion **25**. A material of the third contact portion **173** may be the same as that of the front surface side metal layer **110**. The third contact portion **173** may include a plug member such as tungsten and may include a barrier metal. The third contact portion **173** is provided inside the contact hole **55**.

[0148] FIG. 6B illustrates an example of a cross section e-e' in FIG. 4A. The cross section e-e' is an XZ plane passing through the resistance portion **62** and the connection portion **25**. The cross section e-e' crosses the resistance adjustment portion **60** in the X axis direction and passes through the third contact portion **173**.

[0149] The diode element portion **61** may be provided to cover the resistance portion **62**. A width in the X axis direction of the diode element portion **61** in the present example is larger than a width in

the X axis direction of the resistance portion **62**. A width in the Y axis direction of the diode element portion **61** may be larger or smaller than a width in the Y axis direction of the resistance portion **62**.

[0150] The first polysilicon layer **91** functioning as the connection portion **25** may be provided below the gate metal wiring **152**. The first polysilicon layer **91** functioning as the connection portion **25** is separated from the second polysilicon layer **92** functioning as the diode element portion **61** by the interlayer dielectric film **38**.

[0151] FIG. 7A illustrates an example of a method for manufacturing the semiconductor device **100** including the resistance adjustment portion **60**. In step **S100**, the first insulating portion **81** is formed above the semiconductor substrate **10**.

[0152] In step **S102**, the first polysilicon layer **91** for forming the resistance portion **62** is formed on the first insulating portion **81**. The first polysilicon layer **91** may be patterned according to a shape of the resistance portion **62**. The first polysilicon layer **91** may be formed simultaneously with the gate conductive portion **44** and the gate conductive portion **194**. A third opening **273** may be formed in the first polysilicon layer **91** before a step of forming the second insulating portion **82**. The third opening **273** may be an opening for forming the first contact portion **171** and the third insulating portion **83**. However, the formation of the third opening **273** may be omitted.

[0153] In step **S104**, the second insulating portion **82** is formed on the first polysilicon layer **91**. When the third opening **273** is formed, the third insulating portion **83** may be formed inside the third opening **273**. The third insulating portion **83** may be formed by the same process as that of the second insulating portion **82**. The material of the third insulating portion **83** may be the same as the material of the second insulating portion **82**. When the second insulating portion **82** is formed, a contact hole **57** to be described later may be formed by patterning, etching, or the like. The second polysilicon layer **92** for forming the diode element portion **61** is formed on the second insulating portion **82**. The second polysilicon layer **92** may be formed by the process common with that of the temperature sense diode of the temperature sensing portion **180**. A PN structure of the diode element portion **61** may be formed by a process common with that of a PN structure of the temperature sensing portion **180**. The second polysilicon layer **92** may be formed up to an inside of the contact hole **57**.

[0154] FIG. 7B illustrates an example of the method for manufacturing the semiconductor device **100** including the resistance adjustment portion **60**. This drawing illustrates a step subsequent to the method for manufacturing the semiconductor device **100** of FIG. 7A.

[0155] In step **S106**, the interlayer dielectric film **38** is formed above the diode element portion **61** and the resistance portion **62**. After the interlayer dielectric film **38** is formed, the first opening **271** for forming the first contact portion **171** may be formed to penetrate the interlayer dielectric film **38**, the second polysilicon layer **92**, and the second insulating portion **82**. After the interlayer dielectric film **38** is formed, the second opening **272** for forming the second contact portion **172** may be formed to penetrate the interlayer dielectric film **38**, the second polysilicon layer **92**, the second insulating portion **82**, and the first polysilicon layer **91**. In step **S104**, when the contact hole **57** is formed, the second opening **272** may not be formed.

[0156] The first opening **271** and the second opening **272** may be formed by a same etching process. The first opening **271** and the second opening **272** may reach an upper surface of the first insulating portion **81**. The first opening **271** and the second opening **272** may extend inside the first insulating portion **81** or may not reach the upper surface of the first insulating portion **81**.

[0157] The step of forming the first opening **271** may include a step of forming the first opening **271** to penetrate the third insulating portion **83**. In the step of forming the first opening **271**, in order to separate the first contact portion **171**, which fills the first opening **271**, from the resistance portion **62**, the first opening **271** is formed such that the third insulating portion **83** is provided between the first opening **271** and the first polysilicon layer **91**.

[0158] In step **S108**, the first contact portion **171** is formed in the first opening **271**, and the second

contact portion **172** is formed in the second opening **272**. The first contact portion **171** and the second contact portion **172** may be formed by a same contact forming process. The step of forming the first contact portion **171** may include a step of forming the first contact portion **171** inside the third insulating portion **83**. In step **S106**, when the second opening **272** is not formed, the second contact portion **172** may not be formed.

[0159] In the method for manufacturing the semiconductor device **100** in the present example, the process can be simplified by making at least a part of the step for forming the resistance adjustment portion **60** common with the step for forming the transistor portion **70**, the current sensing portion **90**, or the temperature sensing portion **180**. By making all the steps of the resistance adjustment portion **60** common, a new step for forming the resistance adjustment portion **60** can be made unnecessary.

[0160] FIG. **8** illustrates a modification of the cross section c-c' in FIG. **4A**. Differences from FIG. **5A** will be described with reference to FIG. **8**.

[0161] In the example of FIG. **8**, the diode element portion **61** and the resistance portion **62** are in direct contact with each other via the contact hole **57** provided in the second insulating portion **82**. That is, in the example of FIG. **8**, the second contact portion **172** and the second opening **272** are not provided. By bringing the diode element portion **61** and the resistance portion **62** into direct contact with each other, a size of the contact hole **57** can be adjusted, thereby adjusting characteristics and adjusting the resistance value of the resistance portion **62**.

[0162] Inside the contact hole **57**, a height of an interface between the diode element portion **61** and the resistance portion **62** may be the same as a height of a lower surface of the second insulating portion **82**. As an example, the contact hole **57** is formed by forming the second insulating portion **82** on the first polysilicon layer **91** and then selectively removing the second insulating portion **82** by etching or the like. Note that by performing over-etching when the second insulating portion **82** is removed, the height of the interface between the diode element portion **61** and the resistance portion **62** may be located below the height of the lower surface of the second insulating portion **82**.

[0163] Inside the contact hole **57**, the height of the interface between the diode element portion **61** and the resistance portion **62** may be located above the height of the lower surface of the second insulating portion **82**. That is, the first polysilicon layer **91** may be provided to extend to the inside of the contact hole **57** in the depth direction of the semiconductor substrate **10**. As an example, the contact hole **57** is formed by selectively forming a mask on the first polysilicon layer **91**, etching the first polysilicon layer **91**, then forming the second insulating portion **82** on the first polysilicon layer **91**, and removing the mask.

[0164] FIG. **9A** illustrates a modification of the cross section c-c' in FIG. **4A**. The resistance adjustment portion **60** in the present example is different from the example illustrated in FIG. **5A** or the like in that the resistance adjustment portion **60** includes a resistance adjustment trench portion **65**, which is provided to extend in a predetermined direction, in the semiconductor substrate **10**. Difference from the example of the cross section c-c' illustrated in FIG. **5A** will be described with reference to FIG. **9A**.

[0165] The first polysilicon layer **91** is provided inside the resistance adjustment trench portion **65**. That is, in the example illustrated in FIG. **9A**, the first polysilicon layer **91** is not provided on the first insulating portion **81** formed above the front surface **21** of the semiconductor substrate **10**, but is provided inside the resistance adjustment trench portion **65** which is provided to extend in the depth direction from the front surface **21** of the semiconductor substrate **10**. The first polysilicon layer **91** may function as the resistance portion **62**.

[0166] The resistance adjustment trench portion **65** in the present example includes a resistance adjustment dielectric film **64** and a first polysilicon layer **91**. The resistance adjustment dielectric film **64** may be formed by oxidizing or nitriding a semiconductor on an inner wall of a resistance adjustment trench. The first polysilicon layer **91** is formed farther inward than the resistance

adjustment dielectric film **64** inside the resistance adjustment trench portion **65**. The resistance adjustment dielectric film **64** insulates the first polysilicon layer **91** from the semiconductor substrate **10**.

[0167] The second polysilicon layer **92** is provided on the first insulating portion **81**, the second insulating portion **82**, and the first polysilicon layer **91**. The first insulating portion **81** and the second insulating portion **82** may be formed in a same step or may be formed in different steps. The second polysilicon layer **92** may function as the diode element portion **61**. An interface **93** is formed between the first polysilicon layer **91** and the second polysilicon layer **92**.

[0168] The second polysilicon layer **92** is provided in direct contact with the first polysilicon layer **91**. That is, in the example of FIG. **9A**, the second contact portion **172** and the second opening **272** are not provided. Accordingly, the diode element portion **61** and the resistance portion **62** can be brought into direct contact with each other, and by adjusting a size of the contact interface **93** at the contact, a resistance value of the resistance adjustment portion **60** can be adjusted. Note that in the example of FIG. **9A**, the second contact portion **172** and the second opening **272** may be provided.

[0169] FIG. **9B** illustrates a modification of the cross section c-c' in FIG. **4A**. The example of FIG. **9B** is different from the example illustrated in FIG. **9A** in that the first polysilicon layer **91** provided in the resistance adjustment trench portion **65** functions as a part of the resistance portion **62** and the diode element portion **61**.

[0170] The diode element portion **61** includes a cathode region **361** and an anode region **362**. The cathode region **361** is a region of the first conductivity type. The cathode region **361** in the present example is the N type. The anode region **362** is a region of the second conductivity type. The anode region **362** in the present example is the P type. In the diode element portion **61**, a current flows in a direction from the anode region **362** toward the cathode region **361**.

[0171] At least one of the cathode region **361** or the anode region **362** may be provided in the resistance adjustment trench portion **65**. In the present example, the cathode region **361** is provided in the resistance adjustment trench portion **65** which is provided to extend in a predetermined direction in the semiconductor substrate **10**. The anode region **362** may be provided in the resistance adjustment trench portion **65**.

[0172] In this manner, the first polysilicon layer **91** may function not only as the resistance portion **62** but also as one of the cathode region **361** or the anode region **362** of the diode element portion **61**. Accordingly, by changing the size of the interface **93**, the characteristics of the diode element portion **61** can be adjusted, or the resistance adjustment portion **60** can be downsized.

[0173] FIG. **10** illustrates an example of a top view of the resistance adjustment portion **60**. As an example, FIG. **10** is a top view of the resistance adjustment portion **60** illustrated in FIG. **9A**. Since the resistance portion **62** is provided below the diode element portion **61**, the resistance portion **62** is invisible in top view, but in FIG. **10**, a part provided below the diode element portion **61** is indicated by a dotted line in order to grasp a positional relationship. The resistance adjustment portion **60** in the present example includes the resistance adjustment trench portion **65** which is provided to extend in the predetermined direction in the semiconductor substrate **10**. The resistance portion **62** is provided in the resistance adjustment trench portion **65** in the present example.

[0174] The resistance adjustment trench portion **65** in the present example is provided to extend in the Y axis direction. That is, the resistance adjustment trench portion **65** in the present example is provided to extend in a rectification direction of the diode element portion **61**, but is not limited thereto. The resistance adjustment trench portion **65** may be provided to extend in the X axis direction.

[0175] A plurality of the resistance adjustment trench portions **65** may be provided. The resistance adjustment portion **60** in the present example includes six resistance adjustment trench portions **65**. An end portion of the resistance adjustment trench portion **65** in the present example has a U shape on the front surface **21** of the semiconductor substrate **10**, similarly to the gate trench portion **40**. That is, the resistance adjustment trench portion **65** may include two extending parts extending

along the extending direction and a connecting part connecting the two extending parts.

[0176] A length of the resistance adjustment trench portion **65** in a predetermined direction may be shorter than a length of the gate trench portion **40** in the current sensing portion **90**. The length of the resistance adjustment trench portion **65** may be 2 μm or more and 2000 μm or less. In addition, the length of the resistance adjustment trench portion **65** may be 50 μm or more and 500 μm or less. By adjusting the length of the resistance adjustment trench portion **65**, the resistance value of the resistance adjustment portion **60** can be adjusted.

[0177] The resistance adjustment trench portion **65** may be provided in a shape in which a plurality of U-shaped trenches are connected. The resistance adjustment trench portion **65** may include n extending parts (n is a natural number of 2 or more) extending along the extending direction and n-1 connecting part(s) each connecting two extending parts. With such a shape, the resistance value of the resistance adjustment portion **60** can be easily adjusted.

[0178] The semiconductor device **100** in the present example may include both the resistance adjustment portion **60** including the resistance adjustment trench portion **65** and the resistance adjustment portion **60** not including the resistance adjustment trench portion **65**. That is, the main adjustment portion **160** may include the resistance adjustment trench portion **65**, and the sense adjustment portion **260** may not include the resistance adjustment trench portion **65**. Alternatively, the main adjustment portion **160** may not include the resistance adjustment trench portion **65**, and the sense adjustment portion **260** may include the resistance adjustment trench portion **65**.

[0179] While the present invention has been described by way of the embodiments, the technical scope of the present invention is not limited to the above-described embodiments. It is apparent to persons skilled in the art that various alterations or improvements can be made to the above described embodiments. It is also apparent from the described scope of the claims that the embodiments to which such alterations or improvements are made can be included in the technical scope of the present invention. For example, in the resistance adjustment portion **60**, the resistance portion **62** may be arranged on the gate pad **150** side of the diode element portion **61**.

[0180] The operations, procedures, steps, and stages of each process performed by an apparatus, system, program, and method shown in the claims, embodiments, or diagrams can be performed in any order as long as the order is not indicated by “prior to,” “before,” or the like and as long as the output from a previous process is not used in a later process. Even if the process flow is described using phrases such as “first” or “next” in the claims, the specification, or the drawings for the sake of convenience, it does not necessarily mean that the process must be performed in this order.

Explanation of References

[0181] **10**: semiconductor substrate; **12**: emitter region; **14**: base region; **15**: contact region; **16**: accumulation region; **17**: well region; **18**: drift region; **20**: buffer region; **21**: front surface; **22**: collector region; **23**: back surface; **24**: collector electrode; **25**: connection portion; **30**: dummy trench portion; **32**: dummy dielectric film; **34**: dummy conductive portion; **38**: interlayer dielectric film; **40**: gate trench portion; **41**: extending part; **42**: gate dielectric film; **43**: connecting part; **44**: gate conductive portion; **52**: emitter electrode; **53**: contact hole; **54**: contact hole; **55**: contact hole; **56**: contact hole; **57**: contact hole; **60**: resistance adjustment portion; **61**: diode element portion; **62**: resistance portion; **63**: resistance region; **64**: resistance adjustment dielectric film; **65**: resistance adjustment trench portion; **70**: transistor portion; **71**: mesa portion; **81**: first insulating portion; **82**: second insulating portion; **83**: third insulating portion; **90**: current sensing portion; **91**: first polysilicon layer; **92**: second polysilicon layer; **93**: interface; **100**: semiconductor device; **102**: end side; **104**: active portion; **106**: outer peripheral portion; **110**: front surface side metal layer;

[0183] **120**: back surface side lifetime control region; **130**: anode pad; **132**: anode wiring; **140**: cathode pad; **142**: cathode wiring; **150**: gate pad; **152**: gate metal wiring; **160**: main adjustment portion; **161**: main on-side diode; **162**: main off-side diode; **166**: main on-side resistor; **167**: main off-side resistor; **171**: first contact portion; **172**: second contact portion; **173**: third contact portion;

180: temperature sensing portion; **190**: current sense pad; **194**: gate conductive portion; **260**: sense adjustment portion; **261**: sense on-side diode; **262**: sense off-side diode; **266**: sense on-side resistor; **267**: sense off-side resistor; **271**: first opening; **272**: second opening; **273**: third opening; **361**: cathode region; and **362**: anode region.

Claims

1. A semiconductor device comprising: a transistor portion; a current sensing portion which detects a current flowing through the transistor portion; a gate pad which is provided above a semiconductor substrate; and a resistance adjustment portion which is electrically connected to the gate pad and adjusts gate resistances of the transistor portion and the current sensing portion, wherein the resistance adjustment portion includes a main adjustment portion which is electrically connected to a gate conductive portion of the transistor portion, and a sense adjustment portion which is electrically connected to a gate conductive portion of the current sensing portion, and each of the main adjustment portion and the sense adjustment portion includes a diode element portion including a plurality of diodes provided in anti-parallel and a resistance portion which is connected to the diode element portion.
2. The semiconductor device according to claim 1, wherein the resistance portion has a resistance value such that the transistor portion is turned on before the current sensing portion, and the transistor portion is turned off later than the current sensing portion.
3. The semiconductor device according to claim 1, wherein the main adjustment portion includes a main on-side diode which has an anode electrically connected to the gate pad and a cathode electrically connected to the gate conductive portion of the transistor portion, a main off-side diode which has an anode electrically connected to the gate conductive portion of the transistor portion and a cathode electrically connected to the gate pad, and a main off-side resistor which is electrically connected between the anode of the main off-side diode and the gate conductive portion of the transistor portion.
4. The semiconductor device according to claim 1, wherein the sense adjustment portion includes a sense on-side diode which has an anode electrically connected to the gate pad and a cathode electrically connected to the gate conductive portion of the current sensing portion, a sense off-side diode which has an anode electrically connected to the gate conductive portion of the current sensing portion and a cathode electrically connected to the gate pad, and a sense on-side resistor which is electrically connected between the cathode of the sense on-side diode and the gate conductive portion of the current sensing portion.
5. The semiconductor device according to claim 1, wherein the main adjustment portion includes a main on-side diode which has an anode electrically connected to the gate pad and a cathode electrically connected to the gate conductive portion of the transistor portion, a main off-side diode which has an anode electrically connected to the gate conductive portion of the transistor portion and a cathode electrically connected to the gate pad, a main on-side resistor which is electrically connected between the cathode of the main on-side diode and the gate conductive portion of the transistor portion, and a main off-side resistor which is electrically connected between the anode of the main off-side diode and the gate conductive portion of the transistor portion.
6. The semiconductor device according to claim 5, wherein the sense adjustment portion includes a sense on-side diode which has an anode electrically connected to the gate pad and a cathode electrically connected to the gate conductive portion of the current sensing portion, a sense off-side diode which has an anode electrically connected to the gate conductive portion of the current sensing portion and a cathode electrically connected to the gate pad, a sense on-side resistor which is electrically connected between the cathode of the sense on-side diode and the gate conductive portion of the current sensing portion, and a sense off-side resistor which is electrically connected between the anode of the sense off-side diode and the gate conductive portion of the current

sensing portion.

7. The semiconductor device according to claim 1, wherein in the resistance adjustment portion, the diode element portion is provided above the resistance portion.

8. The semiconductor device according to claim 7, comprising: a first insulating portion which is provided above the semiconductor substrate; a second insulating portion which is provided above the first insulating portion; an interlayer dielectric film which is provided above the first insulating portion and the second insulating portion; and a front surface side metal layer which is provided on the interlayer dielectric film, wherein the resistance portion is provided on the first insulating portion, and the diode element portion is provided above the resistance portion with the second insulating portion interposed therebetween.

9. The semiconductor device according to claim 8, comprising a first contact portion which is provided to extend from an upper end of the interlayer dielectric film to an upper end of the first insulating portion through the diode element portion, the second insulating portion, and the resistance portion, and connects the front surface side metal layer and the diode element portion.

10. The semiconductor device according to claim 9, comprising a third insulating portion which is provided around the first contact portion and electrically separates the first contact portion from the resistance portion.

11. The semiconductor device according to claim 8, comprising a second contact portion which is provided to extend from an upper end of the interlayer dielectric film to an upper end of the first insulating portion through the diode element portion, the second insulating portion, and the resistance portion, and connects the front surface side metal layer and the resistance portion.

12. The semiconductor device according to claim 1, wherein the resistance adjustment portion includes a resistance adjustment trench portion, which is provided to extend in a predetermined direction, in the semiconductor substrate, and the resistance portion is provided in the resistance adjustment trench portion.

13. The semiconductor device according to claim 8, wherein the diode element portion and the resistance portion are in direct contact with each other via a contact hole provided in the second insulating portion.

14. The semiconductor device according to claim 1, wherein the resistance adjustment portion includes a resistance adjustment trench portion, which is provided to extend in a predetermined direction, in the semiconductor substrate, a first polysilicon layer which is provided inside the resistance adjustment trench portion, and a second polysilicon layer which is provided in direct contact with the first polysilicon layer.

15. The semiconductor device according to claim 1, wherein the resistance adjustment portion includes a resistance adjustment trench portion, which is provided to extend in a predetermined direction, in the semiconductor substrate, the diode element portion includes a cathode region and an anode region, and at least one of the cathode region or the anode region is provided in the resistance adjustment trench portion.

16. The semiconductor device according to claim 1, wherein a thickness of the resistance portion is smaller than a thickness of the diode element portion.

17. A method for manufacturing a semiconductor device including a transistor portion and a current sensing portion, the method comprising: providing a gate pad above a semiconductor substrate; and providing a resistance adjustment portion which is electrically connected to the gate pad and adjusts gate resistances of the transistor portion and the current sensing portion, wherein the providing the resistance adjustment portion includes forming a main adjustment portion which is electrically connected to a gate conductive portion of the transistor portion, and forming a sense adjustment portion which is electrically connected to a gate conductive portion of the current sensing portion, and each of the main adjustment portion and the sense adjustment portion includes a diode element portion including a plurality of diodes provided in anti-parallel and a resistance portion which is connected to the diode element portion.

18. The method for manufacturing the semiconductor device according to claim 17, the method comprising: forming a first insulating portion above the semiconductor substrate; forming a first polysilicon layer for forming the resistance portion on the first insulating portion; forming a second insulating portion on the first polysilicon layer; forming a second polysilicon layer for forming the diode element portion on the second insulating portion; forming an interlayer dielectric film above the diode element portion and the resistance portion; forming a first opening, which is for forming a first contact portion, to penetrate the interlayer dielectric film, the second polysilicon layer, and the second insulating portion; forming a second opening, which is for forming a second contact portion, to penetrate the interlayer dielectric film, the second polysilicon layer, the second insulating portion, and the first polysilicon layer; forming the first contact portion in the first opening; and forming the second contact portion in the second opening.

19. The method for manufacturing the semiconductor device according to claim 18, wherein the first opening and the second opening are formed by a same etching process, and the first contact portion and the second contact portion are formed by a same contact forming process.

20. The method for manufacturing a semiconductor device according to claim 17, the method comprising: forming a first insulating portion above the semiconductor substrate; forming a first polysilicon layer for forming the resistance portion on the first insulating portion; forming a second insulating portion on the first polysilicon layer; forming a second polysilicon layer for forming the diode element portion on the second insulating portion; forming an interlayer dielectric film above the diode element portion and the resistance portion; forming a first opening, which is for forming a first contact portion, to penetrate the interlayer dielectric film, the second polysilicon layer, and the second insulating portion; and forming the first contact portion in the first opening.

21. The method for manufacturing the semiconductor device according to claim 18, the method comprising: forming a third opening in the first polysilicon layer before the forming the second insulating portion; and forming a third insulating portion in the third opening, wherein the forming the first opening includes forming the first opening to penetrate the third insulating portion, and the forming the first contact portion includes forming the first contact portion inside the third insulating portion.
